NPTEL ETHICAL HACKING & NETWORK COMMUNICATIONS

WEEK 3 DETAILED CLASSROOM STUDY NOTES & EXAM PREPARATION NOTEBOOK

Course: NPTEL Ethical Hacking | **Coverage:** Lectures 11 to 15 (Complete Week 3 Syllabus)

MODULE 1: LECTURE 11 — ROUTING PROTOCOLS (PART I): FOUNDATIONS & DELIVERY

1.1 Connecting Week 1 & Week 2 to Week 3 Routing Mechanics

Bridge from Prior Weeks: In Week 1, we learned that IP operates as an inherently connectionless datagram protocol at Layer 3. In Week 2, we mastered logical IPv4 addressing, subnetting, and CIDR notation (/M). **Why Week 3 is Needed:** Possessing an IP address and subnet mask is like having a house address in a city—it identifies *who* you are, but does not tell you *how to get there*! Routers act as the intelligent traffic controllers (the 'GPS' of the Internet) that inspect incoming packet destination addresses, perform bitwise logical ANDing with subnet masks, and decide the precise outgoing network interface. Without routing protocols and routing tables, packets would remain trapped inside their local LAN segment.

1.2 Connection Options: Logical TCP Connections over Connectionless IP

What it is: Data communication at Layer 3 (IP) is connectionless, whereas Layer 4 (TCP) provides connection-oriented reliability.
• **Connection-Oriented (TCP):** Establishes a logical session between sender and receiver via a 3-way handshake before transmitting payload data. It guarantees in-order, error-checked delivery using sequence numbers, acknowledgments (ACKs), and retransmissions.
• **Connectionless (IP & UDP):** Treats every data packet (datagram) as an independent entity carrying complete destination addressing information. Packets are forwarded individually without establishing a prior channel or reserving network resources.
Crucial Insight: A TCP 'connection' is purely *logical* (maintained in host memory states). It does **NOT** guarantee a fixed physical path! Because the underlying IP layer is connectionless, different TCP segments belonging to the exact same transfer can follow completely different intermediate physical router paths depending on real-time network congestion.

> ■ **KEY CONCEPT TO REMEMBER**
>
> **Key Takeaway:** TCP provides a reliable, connection-oriented service ON TOP OF an unreliable, connectionless IP foundation. The IP layer routes each packet independently based on current routing table metrics.

1.3 Packet Delivery Options: Direct vs. Indirect Delivery

Packet delivery at the IP layer is divided into two distinct operational paradigms based on whether the source and destination reside on the same physical network or different networks:

1. Direct Delivery:
• **What it is:** Occurs when the source host and destination host (or router) are directly connected to the *same physical network*.
• **How it works:** The source node compares its own Subnet ID with the destination's Subnet ID (Destination IP AND Source Subnet Mask). If they match, the packet is encapsulated directly into a Data Link frame (using ARP to resolve the destination MAC address) and sent over the local media. **No intermediate IP routers are required to cross network boundaries.**
• **Examples:** Host A (192.168.1.10/24) sending to Host B (192.168.1.20/24), or Host A sending directly to its Default Gateway router interface (192.168.1.1).

2. Indirect Delivery:
• **What it is:** Occurs when the source and destination reside on *different networks*.

• **How it works:** The packet must traverse one or more intermediate routers (hops) before reaching the destination network. The source sends the packet to its Default Gateway router via a direct delivery step. The router inspects its routing table and forwards the packet to the next-hop router (indirect step). This indirect forwarding continues hop-by-hop until the packet reaches the final router connected to the target network. The final router then performs a **Direct Delivery** to the destination host!

Delivery Paradigm	Source & Destination Location	Routers Involved?	Final Address Resolution
Direct Delivery	Same physical network segment (LAN)	No intermediate IP routers needed	ARP resolves Destination Host MAC address directly
Indirect Delivery	Different logical networks across WAN/Internet	One or more intermediate routers (hops)	ARP resolves Next-Hop Gateway MAC address at each hop

1.4 Routing Table Types & Entry Classifications

Every router and multi-homed host maintains a **Routing Table**—a specialized data structure acting as a network map. When an IP packet arrives, the router consults this table to make forwarding decisions. Routing tables contain four primary types of entries:
1. **Next-Hop Routing:** Rather than storing full end-to-end paths (which would require infinite memory), the routing table only stores the address of the *immediate next router* (Gateway) or outgoing interface. This simplifies routing calculations drastically.
2. **Network-Specific Routing:** Stores a single route entry for an entire network segment (e.g., 192.168.10.0/24) rather than listing individual host IPs. This reduces table size from millions of entries to a few summarized network entries.
3. **Host-Specific Routing:** Stores an explicit route for a single specific IP address (subnet mask 255.255.255.255 or /32). Used for special diagnostic routes, security gateways, or high-priority servers. Marked by the **'H' flag** in routing tables.
4. **Default Routing (Gateway of Last Resort):** A catch-all route designated by address **0.0.0.0/0** (or mask 0.0.0.0). If an incoming packet's destination does not match any network or host entry in the table, it is forwarded out the default route interface to an ISP or backbone router.

■ ROUTING TABLE ADMINISTRATION

Static vs. Dynamic Routing Tables:
• **Static Routing:** Entries are manually typed into the router by a network administrator. Pros: Simple, secure, zero CPU/bandwidth overhead. Cons: Does not adapt to link failures; requires manual reconfiguration.
• **Dynamic Routing:** Entries are automatically discovered, updated, and re-routed in real-time by routing protocols (e.g., RIP, OSPF, BGP) when network topology or link status changes.

1.5 Inspection Commands & Routing Table Flags

In classroom and lab environments, security analysts inspect local routing tables using command-line interface (CLI) utilities:
• **Unix / Linux:** Executing netstat -r (or ip route show) displays destination networks, gateways, subnet masks, flags, and interface names.
• **Windows:** Executing route print provides IPv4 and IPv6 active routes, gateway addresses, interface indices, and metrics.

Standard Routing Table Flags to Memorize:
• **U (Up):** The route is active and operational.
• **G (Gateway):** The destination is on an external network; indirect routing is required via a gateway router.
• **H (Host):** The route is host-specific (a single host IP address rather than a subnet).
• **D (Dynamic):** Route was dynamically added by a daemon or ICMP redirect message.
• **M (Modified):** Route was dynamically updated or modified by a routing protocol.

■ EXAM POINT / COMMON MISTAKE

Exam Focus: On NPTEL exams, you will be asked to identify flags! Remember: Flag **G** indicates an *indirect route* (gateway required). Flag **H** indicates a *host-specific route*. Default route is always represented as **0.0.0.0 with mask 0.0.0.0**.

MODULE 2: LECTURE 12 — INTERIOR ROUTING PROTOCOLS: RIP & OSPF

2.1 Routing Operational Scope: IGPs, EGPs, and Autonomous Systems (AS)

Routing protocols are broadly classified into two operational domains based on administrative boundaries:
• **Autonomous System (AS):** A collection of interconnected routers and networks managed by a single administrative entity or organization (e.g., a university campus, enterprise corporation, or Internet Service Provider) that presents a unified routing policy to the outside world. Each public AS is assigned a unique 16-bit or 32-bit **AS Number (ASN)** by regional registries (e.g., ARIN, RIPE).
• **Interior Gateway Protocols (IGPs):** Used to exchange routing updates exclusively *within* a single Autonomous System. Examples: RIP, OSPF, IS-IS, EIGRP. Primary goal: Maximum performance, fast convergence, and optimal internal path computation.
• **Exterior Gateway Protocols (EGPs):** Used to route traffic *between* different Autonomous Systems across the global Internet. Primary example: **BGP (Border Gateway Protocol)**. Primary goal: Policy enforcement, administrative reachability, and path vector security.

2.2 Routing Information Protocol (RIP) & Distance Vector Algorithmic Flaws

What it is: RIP is one of the oldest IGPs, employing the **Distance Vector Routing** algorithm based on the Bellman-Ford equation.
How it Works:
1. **Metric:** Measures path cost strictly by **Hop Count** (the number of intermediate routers a packet must cross). Each link has a cost of 1.
2. **Periodic Updates:** Every 30 seconds, each RIP router broadcasts or multicasts its *entire routing table* to directly connected neighbors.
3. **Distance Vector Equation:** If router X receives a vector from neighbor V advertising a route to destination Y at cost D_v(Y), X calculates its new cost as:
   D_x(Y) = min_v { c(x,v) + D_v(Y) }
4. **Hop Count Limit:** The maximum allowable hop count is **15**. A hop count value of **16 represents Infinity (Unreachable)**.

Severe Vulnerabilities & Algorithmic Limitations:
• **Slow Convergence:** Topology changes propagate slowly step-by-step across neighbors.
• **Count-to-Infinity Problem:** When a link fails, neighboring routers can enter a mutual update loop where hop counts increment endlessly (16, 17... capped at 16) before realizing the destination is dead!
• **High Bandwidth Overhead:** Transmitting full routing tables every 30 seconds consumes significant link bandwidth.
• **Mitigation Techniques:** Split Horizon (never advertise a route back out the interface it was learned from), Poison Reverse (advertise dead routes with cost 16), and Hold-down Timers.

> ■ **EXAM POINT / COMMON MISTAKE**
>
> **NPTEL Exam Focus:** RIP uses Hop Count as its metric (Max 15; 16 = Infinity). RIP suffers from slow convergence and the 'Count-to-Infinity' problem caused by routing loops during link failures.

2.3 Open Shortest Path First (OSPF) & Link-State Architecture

What it is: OSPF is an advanced, widely used open-standard IGP based on the **Link-State Routing** paradigm.
Why it is Superior: Unlike RIP which shares unverified rumors ('entire tables to neighbors'), OSPF routers share factual local status reports ('my directly connected link states to everyone'). Every router builds a complete, identical map of the network topology and independently computes the absolute shortest paths using **Dijkstra's Shortest Path First (SPF) Algorithm**.

Step-by-Step OSPF Operation:
1. **Neighbor Discovery:** Routers send small **Hello Packets** periodically (default 10s) out all active interfaces to discover neighbors and verify bidirectional liveness.
2. **LSA Flooding:** Each router generates **Link State Advertisements (LSAs)** describing its active links, interface costs, and

connected neighbors. LSAs are flooded throughout the OSPF Area.
3. **LSDB Construction:** Every router compiles incoming LSAs into an identical **Link State Database (LSDB)** representing the network graph.
4. **Dijkstra Execution:** Each router executes Dijkstra's algorithm placing itself as the root vertex to compute the shortest path tree to all destinations.
5. **Routing Table Population:** The lowest-cost next-hop entries are copied into the IP Routing Table.

OSPF Metric (Cost Formula):
OSPF calculates link cost based on bandwidth: **Cost = Reference Bandwidth / Interface Bandwidth**.
Default Reference Bandwidth = 10^8 bps (100 Mbps). E.g., a 10 Mbps link has Cost = 10^8 / 10^7 = 10; a 100 Mbps link has Cost = 1.

2.4 OSPF Packet Types & Header Structure

OSPF operates directly on top of IP using **IP Protocol Number 89** (no TCP/UDP overhead). It defines 5 distinct message types:

Type #	OSPF Message Name	Primary Functional Purpose in OSPF Lifecycle
Type 1	Hello Packet	Establishes/maintains neighbor relationships; checks liveness (40s Dead Interval)
Type 2	Database Description (DBD)	Summarizes LSDB contents during initial neighbor synchronization
Type 3	Link State Request (LSR)	Requests specific detailed LSAs missing from a router's database
Type 4	Link State Update (LSU)	Carries one or more detailed LSAs; flooded during topology changes
Type 5	Link State Acknowledgment (LSAck)	Confirms reliable receipt of LSU packets (ensures zero data loss)

OSPF Header Fields & Security Authentication:
The 24-byte common OSPF header contains: **Version Number** (v2 for IPv4, v3 for IPv6), **Type** (1-5), **Packet Length**, **Router ID** (32-bit IP identifying the sending router), **Area ID** (32-bit area identifier), **Checksum**, **Authentication Type** (0 = None, 1 = Simple Cleartext Password, 2 = Cryptographic MD5/HMAC), and **Authentication Data**.
Ethical Hacking / Security Relevance: If OSPF authentication is disabled (Type 0) or cleartext (Type 1), an attacker connected to a local switch can inject malicious LSU packets advertising a fake zero-cost link to a rogue gateway, poisoning all router tables in the AS and performing a full network Man-In-The-Middle (MITM) or Denial-of-Service (DoS) attack!

> ■ **KEY CONCEPT TO REMEMBER**
>
> **OSPF Summary:** OSPF uses Link-State Advertisements (LSAs), Dijkstra's Algorithm, Hello Packets (Type 1), and IP Protocol 89. It supports cryptographic authentication to protect against routing table injection attacks.

MODULE 3: LECTURE 13 — EXTERIOR ROUTING PROTOCOLS: BGP ARCHITECTURE

3.1 Border Gateway Protocol (BGP) & Path-Vector Routing Paradigm

What it is: BGP (specifically BGP-4) is the standard Exterior Gateway Protocol (EGP) that forms the backbone of global Internet routing, enabling border routers across distinct Autonomous Systems (AS) to exchange reachability information.
Why it is Essential: Interior protocols (RIP/OSPF) cannot scale to millions of global networks and cannot enforce political or commercial policy decisions. BGP acts as the 'diplomat of the Internet,' allowing independent organizations (ISPs, Tech Giants, Governments) to negotiate traffic routes based on business contracts.

Path-Vector Protocol Mechanics:
• BGP is neither distance-vector nor link-state; it is a **Path-Vector Protocol**.
• Instead of advertising a hop count or link cost, a BGP router advertises a destination prefix accompanied by the **complete list of Autonomous Systems (AS Path)** that traffic must cross to reach that destination (e.g., Destination: 203.0.113.0/24 | AS Path: [AS 100, AS 200, AS 300]).
• **Loop Prevention Guarantee:** If a border router receives a BGP Update whose AS Path contains its own AS Number, it immediately discards the update! This completely eliminates inter-domain routing loops.
• **Transport Connection:** BGP does NOT run directly over IP or Ethernet. BGP establishes reliable, point-to-point sessions over **TCP Port 179**, ensuring guaranteed, error-checked delivery of routing updates.

3.2 BGP Session Lifecycle & The 4 Core Message Types

BGP communication between two peer routers (BGP Peers / Gateways) relies on four standardized message types:

BGP Message Type	Transport / Trigger	Primary Functional Purpose
Open Message	TCP Port 179 (Initial)	Initiates BGP peer session; negotiates protocol parameters, hold times, and ASN
Update Message	TCP Port 179 (Event-driven)	Transmits new reachable routes, modified AS Paths, or withdrawn (infeasible) routes
Keep Alive Message	TCP Port 179 (Periodic 60s)	Maintains peer session active; prevents Hold Timer expiration (default 180s timeout)
Notification Message	TCP Port 179 (Error/Close)	Sent when an error is detected or session is closed; immediately terminates TCP peer link

3.3 Operational Procedures, EBGP vs. IBGP, and BGP Errors

BGP Operational Procedures:
1. **Neighbor Acquisition:** BGP peers establish a TCP connection on Port 179 and exchange Open Messages.
2. **Neighbor Reachability:** Peers exchange Keep Alive Messages every 60 seconds. If no Keep Alive or Update is received within 180 seconds (Hold Time), a Notification is logged and the session is torn down.
3. **Network Reachability:** Peers exchange full routing tables initially, then transmit *incremental Update Messages* only when network topology changes.

External BGP (EBGP) vs. Internal BGP (IBGP):
• **EBGP:** Runs between BGP peers located in *different Autonomous Systems* (different ASNs). Used for inter-AS transit.
• **IBGP:** Runs between BGP peers located *inside the same Autonomous System*. Used to distribute external BGP routes consistently to all internal core routers.

Common Notification Error Conditions:

• **Message Header Error:** Malformed packet length, bad marker field, or syntax corruption.
• **Open Message Error:** Unsupported BGP version, ASN mismatch, or unacceptable hold time.
• **Update Message Error:** Invalid AS Path, missing mandatory attributes, or unresolvable prefix.
• **Hold Timer Expired:** Peer failed to send Keep Alive within 180s.
• **Cease:** Administrative shutdown or deliberate session termination by operator.

3.4 Private IP Address Filtering & BGP Route Hijacking Security Impact

Private IP Addressing in Global Routing:
RFC 1918 designates private IP address blocks (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) for internal organization usage. These addresses are **non-routable on the public Internet**. Public BGP border routers automatically discard any packet bearing a private destination IP address to prevent routing collisions and internal leakage.

Ethical Hacking & Security Impact — BGP Route Hijacking:
BGP was designed assuming mutual trust between network operators and lacks cryptographic origin validation by default. In a **BGP Route Hijacking Attack**, a rogue AS advertises an unauthorized, highly specific IP prefix (e.g., advertising a /24 prefix for a target bank's /22 range). Because routers prioritize the **Longest Prefix Match (LPM)**, global Internet traffic destined for the target is rerouted to the malicious AS! Attackers can inspect, modify, or drop traffic (MITM/DoS). Defense: Modern deployments enforce **RPKI (Resource Public Key Infrastructure)** and BGPsec to validate prefix ownership.

> ■ **EXAM POINT / COMMON MISTAKE**
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> **NPTEL Exam Focus:** BGP is a Path-Vector protocol operating over **TCP Port 179**. Initial exchange sends full tables; subsequent updates are *incremental*. Loop prevention is achieved because routers discard updates containing their own AS Number in the AS Path.

MODULE 4: LECTURE 14 — IP VERSION 6 (IPv6) ARCHITECTURE & TRANSITION

4.1 Drivers for IPv6 & Core Principles Retained from IPv4

The Imperative for IPv6 (IP Next Generation - IPng):
1. **32-Bit Address Space Exhaustion:** IPv4 provides ~4.3 billion addresses. The explosion of mobile devices, cloud infrastructure, and the Internet of Things (IoT) exhausted IPv4 addresses globally.
2. **QoS Limitations in IPv4:** Real-time video conferencing, VoIP, and gaming require guaranteed low latency. IPv4's dynamic packet routing causes variable delays.
3. **Addressing Complexity:** Workarounds like NAT (Network Address Translation) break end-to-end peer-to-peer connectivity and complicate security protocols.

Core Principles Retained from IPv4:
• IPv6 remains a **connectionless datagram protocol** at Layer 3.
• Retains an 8-bit loop prevention counter named **Hop Limit** (functionally identical to IPv4's Time-To-Live / TTL). Decremented by 1 at each router hop; packet is discarded when Hop Limit reaches 0.

4.2 Revolutionary IPv6 Features & 40-Byte Base Header Analysis

Breakthrough Architectural Innovations:
• **128-Bit Address Space:** Provides 2^128 (approx 3.4 x 10^38) unique addresses (~6 x 10^23 addresses per sq. meter of Earth!).
• **Fixed 40-Byte Base Header:** Intermediate routers process headers in hardware at wire speed because base header size is constant (no variable options in base header!).
• **Chained Extension Headers:** Optional features (routing, fragmentation, security) are added as modular extension headers linked via a **Next Header** pointer field.
• **Enhanced QoS (Flow Label & Traffic Class):** A 20-bit **Flow Label** allows a source and routers to bind a stream of packets to a high-speed dedicated path.
• **New Addressing Modes:** Unicast (one-to-one), Multicast (one-to-many; replaces broadcast!), and **Anycast** (one-to-nearest node in a group for DNS/CDN load balancing).
• **Source-Only Fragmentation:** Intermediate routers *NEVER fragment* IPv6 packets! Path MTU Discovery requires the source host to fragment packets using a Fragment Extension Header.

IPv6 Base Field	Size (Bits)	Primary Functional Description
Version	4 bits	Protocol version; hardcoded to binary 0110 (6)
Traffic Class (Priority)	8 bits	Specifies packet class of service for router queue prioritization
Flow Label	20 bits	Identifies specific real-time packet flow requiring QoS path guarantees
Payload Length	16 bits	Length of data payload following base header (extension headers + L4 payload)
Next Header	8 bits	Pointer specifying the type of immediate next header (chained extension or L4 protocol)
Hop Limit	8 bits	Loop prevention counter; decremented by 1 per hop; discards packet at 0
Source Address	128 bits (16B)	128-bit IPv6 address of originating sender host
Destination Address	128 bits (16B)	128-bit IPv6 address of target recipient host

4.3 Colon-Hexadecimal Notation & Zero Compression Rules

Notation Format: 128-bit IPv6 addresses are written as 8 blocks of 4 hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

Zero Compression Rules to Memorize:
1. **Leading Zero Omission:** Leading zeros inside a 16-bit block can be omitted (e.g., :0045: becomes :45:; :0000: becomes :0:).
2. **Double Colon Compression (::):** A contiguous sequence of blocks containing all zeros can be replaced by a single double colon ::.
CRITICAL RULE: The double colon :: can be used **EXACTLY ONCE** in any IPv6 address! Using it twice creates structural ambiguity because a parser cannot determine how many zero blocks belong to each ::.

Worked Compression Example:
Uncompressed: FECD:0000:0000:0000:0001:0000:0000:0023
Step 1 (Omit leading zeros): FECD:0:0:0:1:0:0:23
Step 2 (Compress longest zero sequence): FECD::1:0:0:23 (Notice the second zero sequence 0:0 is left as is because :: was used once!).

4.4 Transition Mechanisms: Dual Stack, Tunneling, and Mapped Addresses

Because IPv4 and IPv6 headers are fundamentally incompatible, the Internet relies on three primary transition mechanisms:
1. **Dual Stack:** Devices run both IPv4 and IPv6 protocol stacks simultaneously on network interfaces, selecting the appropriate stack based on DNS lookup results.
2. **Tunneling:** IPv6 datagrams are encapsulated inside IPv4 packets (IPv4 Header + IPv6 Header + Payload) to cross legacy IPv4 transit networks. **IPv4-Compatible IPv6 Address Format:** 96 bits of zeros + 32-bit IPv4 address (e.g., ::192.168.1.1).
3. **IPv4-Mapped IPv6 Addresses:** Used when an IPv6-enabled host communicates with an IPv4-only host. Format: 80 bits of zeros + 16 bits of ones (FFFF) + 32-bit IPv4 address (e.g., ::ffff:192.168.1.1).
4. **Header Translation:** Converts IPv6 headers directly to IPv4 headers at border gateways (loses Flow Label and extension features).

> ■ **EXAM POINT / COMMON MISTAKE**
>
> **NPTEL Exam Focus:** The IPv6 Base Header size is fixed at **40 bytes**. The **Hop Limit** field is 8 bits (replaces TTL). The double colon :: can be used **only once** per address. IPv4-Mapped addresses have the form ::ffff:x.x.x.x.

MODULE 5: LECTURE 15 — PACKET FORWARDING, LPM RULE & ROUTING SECURITY

5.1 Bitwise Logical ANDing & Table Matching Algorithm

Step-by-Step Packet Forwarding Algorithm inside a Router:
When an IP packet arrives at a router interface, the routing engine executes the following deterministic steps:
1. **Extract Destination IP:** The router reads the 32-bit (IPv4) or 128-bit (IPv6) Destination Address from the IP header.
2. **Iterate Routing Table Rows:** For each row entry in the routing table, the router takes the row's Subnet Mask.
3. **Execute Bitwise ANDing:** Performs a bit-by-bit logical AND operation between the Destination IP and the Subnet Mask:
   Extracted Subnet ID = Destination IP & Row Subnet Mask
4. **Compare Network Addresses:** Compares Extracted Subnet ID with the row's Destination Network Address.
5. **Record Matches & Apply Longest Prefix Match (LPM):** Collects all matching rows and selects the entry with the largest prefix length (/M).
6. **Forward Packet:** Transmits the packet out the designated Gateway / Outgoing Interface.

5.2 Longest Prefix Match (LPM) Rule & Worked Numerical Analysis

What is Longest Prefix Match (LPM)?
When an incoming packet's destination IP address matches *multiple entries* in a routing table, the router MUST select the entry with the longest matching network prefix (the subnet mask containing the highest number of 1s / largest prefix length /M).
Why? A longer prefix represents a smaller, more specific network segment (e.g., /28 is more specific than /24).

WORKED LECTURE EXAMPLE 1: Dual-Match Disambiguation
• Incoming Packet Destination IP: 128.35.57.16
• Routing Table Entry 1: Network 128.35.57.0 | Mask 255.255.255.0 (/24) | Interface eth0
• Routing Table Entry 2: Network 128.35.57.0 | Mask 255.255.255.128 (/25) | Interface eth1

Calculation Step 1 (Row 1 Check):
128.35.57.16 AND 255.255.255.0 = 128.35.57.0 (Matches Row 1 Network!).
Calculation Step 2 (Row 2 Check):
128.35.57.16 AND 255.255.255.128:
Last octet: 16 (00010000) AND 128 (10000000) = 0 (00000000). Result = 128.35.57.0 (Matches Row 2 Network!).
LPM Decision: Row 1 has prefix length /24 (24 bits). Row 2 has prefix length /25 (25 bits). Since 25 > 24, Row 2 is selected! **Packet is forwarded out eth1.**

WORKED LECTURE EXAMPLE 2: Interface Selection in Subnetted Network (/26 = 255.255.255.192)

Incoming Packet IP	Subnet Mask /26 (192 in binary)	Bitwise AND Result (Last Octet)	Matching Subnet	Outgoing Interface
215.1.2.33	33 (00100001) AND 192 (11000000)	0 (00000000)	215.1.2.0/26 (Subnet A: .0-.63)	Interface a
215.1.2.78	78 (01001110) AND 192 (11000000)	64 (01000000)	215.1.2.64/26 (Subnet B: .64-.127)	Interface b
215.1.2.144	144 (10010000) AND 192 (11000000)	128 (10000000)	215.1.2.128/26 (Subnet C: .128-.191)	Interface c
215.1.2.200	200 (11001000) AND 192 (11000000)	192 (11000000)	215.1.2.192/26 (Subnet D: .192-.255)	Interface d

5.3 Table Lookup Optimization & Security Impact of Poisoned Routing

Optimizing Routing Table Search Performance:
Searching millions of routing rows sequentially is computationally expensive. To optimize performance, modern routers sort their routing tables in **descending order of prefix length** (longest prefixes first: /32, /30, /28 ... down to /0 default route). **Optimization Result:** The *very first match* encountered during top-to-bottom lookup is guaranteed to be the Longest Prefix Match! The lookup terminates immediately, saving CPU search time.

Ethical Hacking & Security Relevance — Routing Table Poisoning & Blackholing:
Routing tables dictate traffic flow across the Internet. If an ethical hacker or pen-tester identifies an unauthenticated router, they can demonstrate severe security vulnerabilities:
• **More-Specific Route Injection (MITM):** If a target network uses 10.0.0.0/16, an attacker can inject a 10.0.5.0/24 route. Due to the LPM rule, ALL traffic for 10.0.5.x will bypass the legit /16 route and route to the attacker's machine for interception!
• **Blackhole Attack (DoS):** Injecting a fake route pointing to a non-existent gateway (or null0 interface) drops all enterprise traffic, creating a instant Denial of Service.

> ■ **EXAM POINT / COMMON MISTAKE**
>
> **Exam Focus:** Longest Prefix Match (LPM) selects the route with the highest number of network bits (/M). Sorting routing tables by prefix length (longest first) speeds up forwarding by enabling early exit on first match.

MODULE 6: WEEK 3 REVISION & NPTEL EXAM PREPARATION PAGES

6.1 Quick Week 3 Core Revision Summary

Core Takeaways to Memorize:

• **Connectionless IP:** IP routes datagrams independently. TCP provides logical, reliable connection-oriented delivery on top.
• **Direct vs. Indirect Delivery:** Direct occurs on the same LAN (no router needed to cross net boundary). Indirect traverses routers.
• **Routing Table Entries:** Next-hop, network-specific, host-specific (H flag, /32), and default route (0.0.0.0/0).
• **IGPs vs. EGPs:** IGPs (RIP, OSPF) route within an Autonomous System (AS). EGPs (BGP) route between Autonomous Systems.
• **RIP (Distance Vector):** Hop count metric (Max 15; 16 = Infinity). Suffers from slow convergence and Count-to-Infinity.
• **OSPF (Link-State):** Cost metric based on bandwidth. Uses Hello packets (Type 1, 10s period), LSAs, LSDB, Dijkstra's algorithm, and IP Protocol 89.
• **BGP (Path-Vector):** Runs over TCP Port 179. Advertises AS Paths for loop prevention. Messages: Open, Update, Keep Alive, Notification.
• **IPv6 Architecture:** 128-bit addresses (2^128). Fixed 40-byte base header. Chained extension headers. Hop Limit replaces TTL. Anycast mode added.
• **IPv6 Notation:** Colon-hexadecimal. Double colon :: replaces zero blocks (used EXACTLY ONCE per address!).
• **IPv6 Transition:** Dual Stack (both protocols active), Tunneling (IPv6 in IPv4, ::192.168.1.1), Mapped (::ffff:192.168.1.1).
• **Packet Forwarding & LPM:** Bitwise ANDing extracts Subnet ID. Longest Prefix Match (LPM) selects the most specific route (/M).

6.2 Master Glossary & Important Definitions

Term / Acronym	Technical Definition	Exam & Security Significance
Autonomous System (AS)	Collection of IP routing prefixes under single administrative control with a unique ASN	Defines administrative boundary for routing policy; IGPs run inside, EGPs run outside
Direct Delivery	Packet delivery where source and destination share the same physical network segment	Does not pass through intermediate routers; uses ARP directly for MAC resolution
Indirect Delivery	Packet delivery where source and destination reside on different logical networks	Traverses hop-by-hop through intermediate routers until reaching destination LAN
Longest Prefix Match (LPM)	Rule selecting the routing table entry with the largest prefix length (/M) for a destination	Disambiguates multiple matching routes; exploited in route injection attacks
Distance Vector	Routing algorithm where routers periodically share full tables with direct neighbors	Algorithmic basis of RIP; vulnerable to slow convergence and counting-to-infinity
Link State (LSA)	Routing paradigm where routers flood local link states; all nodes build identical maps	Algorithmic basis of OSPF; fast convergence, uses Dijkstra's shortest path tree
Path Vector (AS Path)	Routing mechanism where updates carry the explicit sequence of AS Numbers traversed	Algorithmic basis of BGP; loop prevention achieved by discarding updates with own ASN
Anycast	IPv6 addressing mode where a packet is delivered to the topologically nearest receiver in a group	Used extensively for global DNS root servers and Content Delivery Networks (CDNs)
Dual Stack	Transition strategy where devices run IPv4 and IPv6 protocol stacks simultaneously	Enables seamless interaction with legacy IPv4 and modern IPv6 destinations
BGP Route Hijacking	Rogue AS advertises an unauthorized, highly specific IP prefix to divert global Internet traffic	Major global threat; forces traffic through attacker AS for MITM interception or DoS

6.3 Important Master Comparisons

Comparison Feature	RIP (Distance Vector)	OSPF (Link-State)	BGP (Path-Vector)
Operational Scope	Interior Gateway Protocol (IGP)	Interior Gateway Protocol (IGP)	Exterior Gateway Protocol (EGP)
Primary Metric	Hop Count (Max 15; 16=Infinity)	Cost (10^8 / Bandwidth in bps)	AS Path & Administrative Policies
Algorithm	Bellman-Ford Distance Vector	Dijkstra's Shortest Path First	Path Vector (AS Path Inspection)
Transport Protocol	UDP Port 520	Directly over IP (Protocol 89)	TCP Port 179
Routing Information Shared	Entire routing table to neighbors	Local link states (LSAs) to area	Reachable prefixes + AS Paths
Convergence Speed	Very Slow (Count-to-Infinity risk)	Very Fast (Immediate LSA flood)	Controlled / Policy-driven
Security / Authentication	Simple Password / MD5 (RIPv2)	Type 0 (None), 1 (Clear), 2 (Crypto/MD5)	RPKI Prefix Validation / BGPsec

6.4 Master Formula Sheet & Important Solved Examples

Master Equations to Memorize:

1. **Bitwise Subnet ID Extraction:** Subnet ID = Destination IP AND Subnet Mask
2. **Bellman-Ford Equation (RIP):** D_x(Y) = min_v { c(x,v) + D_v(Y) }
3. **OSPF Link Cost:** Cost = 10^8 / Interface Bandwidth (bps)
4. **IPv4 Usable Hosts:** Usable Hosts = 2^(32 - Prefix) - 2
5. **IPv6 Double Colon Rule:** :: replaces contiguous zero blocks **EXACTLY ONCE**.

SOLVED EXAMPLE 1: Longest Prefix Match Disambiguation

Problem: A router receives an IP packet destined for 172.16.34.18. Route A: 172.16.32.0/20 via Interface 1 | Route B: 172.16.34.0/24 via Interface 2. Which interface is chosen?

Step 1: Check Route A: 172.16.34.18 AND /20 = 172.16.32.0 (Match!).

Step 2: Check Route B: 172.16.34.18 AND /24 = 172.16.34.0 (Match!).

Step 3: Compare prefix lengths: Route A is /20 (20 bits); Route B is /24 (24 bits).

Conclusion: **Route B (/24) is selected due to Longest Prefix Match. Packet exits Interface 2.**

SOLVED EXAMPLE 2: IPv6 Address Compression & Expansion

Problem: Compress 2001:0DB8:0000:0000:0008:0000:0000:0001.

Step 1 (Omit leading zeros): 2001:DB8:0:0:8:0:0:1

Step 2 (Compress longest zero sequence): The first zero block has 2 segments (0:0); the second zero block has 2 segments (0:0). Compress the first block with ::.

Final Compressed Result: 2001:DB8::8:0:0:1.

6.5 NPTEL Exam Practice MCQs (With Solutions)

Q1: Which statement(s) is/are TRUE regarding connectionless routing used by the IP protocol? (Select all that apply)

a) Each packet is treated as an independent unit for routing decisions

b) A fixed path is established before any packet is transmitted

c) Different packets between the same source and destination may follow different paths

d) IP protocol uses connection-oriented routing

Correct Answer: a, c | *Explanation:* IP uses connectionless datagram routing where packets are routed independently and may take different paths. (Assignment 3 Q1)

Q2: Which statement is FALSE regarding direct and indirect packet delivery?
a) Direct delivery occurs when destination host and deliverer are on the same physical network
b) Indirect delivery occurs when destination host is not on the same network
c) In direct delivery, packets always pass through at least one router
d) In indirect delivery, the packet is routed hop by hop until it reaches the destination network router
Correct Answer: c | *Explanation:* Direct delivery occurs on the same physical network (LAN) and does NOT require passing through a router! (Assignment 3 Q2)

Q3: Which routing table flag indicates that a route is specific to a single host rather than an entire network?
a) U b) G c) H d) D
Correct Answer: c | *Explanation:* Flag 'H' designates a host-specific route (destination is a single host IP with mask 255.255.255.255). (Assignment 3 Q3)

Q4: In the OSPF routing protocol, which type of packet is used to determine whether a neighboring router is active?
a) Link State Update b) Link State Request c) Hello packet d) Link State Acknowledgment
Correct Answer: c | *Explanation:* OSPF Hello packets (Type 1) are exchanged periodically (default 10s) to discover neighbors and verify liveness. (Assignment 3 Q4)

Q5: Which type of address should be used when a packet needs to be delivered to every host within a specific network?
a) Unicast address b) Broadcast address c) Anycast address d) None of these
Correct Answer: b | *Explanation:* Broadcast addresses target all hosts within a specific network segment. (Assignment 3 Q5)

Q6: Which of the following statement(s) is/are FALSE regarding routing protocols?
a) RIP uses distance vector routing and can suffer from count-to-infinity
b) OSPF is a link-state routing protocol
c) BGP is used for routing within a single autonomous system
d) OSPF uses Dijkstra's algorithm to compute shortest paths
e) RIP is typically used for routing between different autonomous systems
Correct Answer: c, e | *Explanation:* BGP is an Exterior protocol (inter-AS), while RIP is an Interior protocol (intra-AS). Thus c and e are false. (Assignment 3 Q6)

Q7: What is the address used to represent the default route in IPv4 routing tables?
a) 127.0.0.1 b) 255.255.255.255 c) 0.0.0.0 d) 192.0.0.1
Correct Answer: c | *Explanation:* The default route (gateway of last resort) is represented by 0.0.0.0 with subnet mask 0.0.0.0 (or 0.0.0.0/0). (Assignment 3 Q7)

Q8: Which statement(s) is/are FALSE regarding IPv6?
a) IPv6 supports address classes similar to IPv4 (Class A, B, C)
b) IPv6 addresses are typically represented in colon-hexadecimal notation
c) IPv6 eliminates fragmentation by intermediate routers
d) The base header size of IPv6 is variable, ranging from 20 to 60 bytes
Correct Answer: a, d | *Explanation:* IPv6 is classless (no Class A/B/C) and has a FIXED 40-byte base header (not 20-60 bytes). Thus a and d are false. (Assignment 3 Q8)

Q9: In the IPv6 base header, how many bits are used for the Hop Limit field?
a) 4 bits b) 8 bits c) 16 bits d) 32 bits
Correct Answer: b | *Explanation:* The Hop Limit field is 8 bits wide (functionally replacing IPv4's 8-bit TTL field). (Assignment 3 Q9)

Q10: An organization is assigned 172.20.64.0/20 network block. How many valid host addresses are available?
a) 2046 b) 4094 c) 4096 d) 8190
Correct Answer: b | *Explanation:* Host bits H = 32 - 20 = 12 bits. Total addresses = 2^12 = 4096. Usable hosts = 4096 - 2 = 4094. (Assignment 3 Q10)

Q11: What underlying transport protocol and port number are used by BGP to establish peer sessions?
a) UDP Port 520 b) TCP Port 179 c) IP Protocol 89 d) TCP Port 80

Correct Answer: b | *Explanation:* BGP operates over TCP Port 179 for reliable inter-AS communication.

Q12: How many times can the double colon (::) zero compression operator be used in a single IPv6 address?
a) Exactly once b) Up to twice c) Unlimited d) Thrice
Correct Answer: a | *Explanation:* The double colon :: can be used EXACTLY ONCE per address to prevent parsing ambiguity.

Q13: Which rule does a router execute when multiple routing table rows match an incoming packet's destination IP?
a) Shortest Hop Count b) First Match Rule c) Longest Prefix Match (LPM) d) Lowest Interface Metric
Correct Answer: c | *Explanation:* Longest Prefix Match (LPM) selects the entry with the highest number of network mask bits (/M).

Q14: What is the primary operational advantage of BGP's Path-Vector mechanism?
a) Eliminates TCP handshakes b) Prevents routing loops via AS Path inspection c) Reduces header size to 20 bytes d) Replaces MAC addresses
Correct Answer: b | *Explanation:* BGP routers drop updates containing their own AS number in the AS Path, preventing inter-domain loops.

Q15: In an IPv6 transition scenario, what is the format of an IPv4-Mapped IPv6 address for host 192.168.1.1?
a) ::192.168.1.1 b) ::ffff:192.168.1.1 c) ffff::192.168.1.1 d) 2001::192.168.1.1
Correct Answer: b | *Explanation:* IPv4-mapped IPv6 addresses take the form ::ffff:w.x.y.z (80 zero bits + 16 one bits + 32-bit IPv4 address).

6.6 Viva Questions & Easy Oral Answers

V1: What is an Autonomous System (AS)?
Answer: An AS is a collection of connected IP routing prefixes under a single administrative control identified by a unique AS Number (ASN).

V2: Differentiate between Direct and Indirect delivery.
Answer: Direct delivery occurs on the same physical network without passing through a router. Indirect delivery traverses intermediate routers to cross network boundaries.

V3: What is the function of the Default Route (0.0.0.0/0)?
Answer: Acts as the 'gateway of last resort' to forward packets whose destinations do not match any specific entry in the routing table.

V4: Why does RIP suffer from the Count-to-Infinity problem?
Answer: Because RIP uses distance vector rumors with slow periodic updates; when a link fails, routers can enter a mutual update loop before marking the route dead at hop count 16.

V5: What algorithm does OSPF use to build its routing table?
Answer: Dijkstra's Shortest Path First (SPF) algorithm executed on an identical Link State Database (LSDB).

V6: Why does BGP use TCP Port 179?
Answer: BGP handles critical global Internet routing and relies on TCP to guarantee ordered, error-checked, and reliable message delivery.

V7: How does BGP prevent inter-domain routing loops?
Answer: Through its Path-Vector mechanism: a router drops any update if its own Autonomous System Number (ASN) appears in the received AS Path.

V8: Why is the IPv6 Base Header fixed at 40 bytes?
Answer: To allow intermediate hardware routers to process packet headers at wire speed without parsing variable IP options.

V9: What is Anycast addressing in IPv6?
Answer: An addressing mode where a packet is delivered to the topologically nearest receiving node among a group sharing the

same Anycast IP.

V10: Why can double colon (::) be used only ONCE in an IPv6 address?
Answer: Because using it twice creates mathematical ambiguity when determining how many zero blocks belong to each :: position.

V11: What is the Longest Prefix Match (LPM) rule?
Answer: A rule stating that when multiple table entries match a destination IP, the router selects the entry with the longest network mask (/M).

V12: Differentiate Dual Stack from Tunneling.
Answer: Dual Stack runs IPv4 and IPv6 simultaneously on a device. Tunneling encapsulates IPv6 packets inside IPv4 headers to cross legacy IPv4 networks.

V13: What is BGP Route Hijacking?
Answer: An attack where a rogue AS advertises an unauthorized, highly specific prefix to illegally reroute global Internet traffic through its network.

V14: What is the metric used by RIP vs OSPF?
Answer: RIP uses Hop Count (Max 15). OSPF uses Cost inversely proportional to bandwidth (10^8 / Bandwidth).

V15: Why don't intermediate routers fragment IPv6 packets?
Answer: IPv6 forces Path MTU Discovery and requires the originating source host to perform all fragmentation using a Fragment Extension Header.

6.7 Final Revision Checklist

Before appearing for the NPTEL Week 3 Exam, verify you can:
[ ] Explain Direct vs. Indirect delivery and perform ARP resolution steps.
[ ] Identify routing table flags (U, G, H, D, M) and inspect tables using netstat -r / route print.
[ ] Explain RIP Distance Vector mechanics, hop count limits (15/16), and the Count-to-Infinity flaw.
[ ] Detail OSPF Link-State operation, Hello packets (Type 1), LSAs, Dijkstra's algorithm, and OSPF header fields.
[ ] Describe BGP Path-Vector operation, AS Paths, TCP Port 179, and the 4 BGP message types.
[ ] Differentiate EBGP from IBGP and explain BGP Route Hijacking security threats.
[ ] List IPv6 architectural features: 128-bit addresses, fixed 40-byte base header, Hop Limit, Anycast mode.
[ ] Master IPv6 Colon-Hexadecimal notation and apply the single :: zero compression rule.
[ ] Differentiate IPv6 transition mechanisms: Dual Stack, Tunneling (::192.168.1.1), and Mapped (::ffff:192.168.1.1).
[ ] Execute bitwise logical ANDing to match destination IPs against subnet masks.
[ ] Apply the Longest Prefix Match (LPM) rule to resolve multi-match forwarding scenarios.